

1. Time Complexity Analysis

Find the time complexity of the following code:

```
c
for(i=1; i<=n; i++)
{
    for(j=1; j<=i; j++)
        printf("*");
}
```

****Calculate:****

- * Number of iterations
- * Big-O notation

2. Binary Search Comparisons

A sorted array contains ****1024 elements****.

Calculate:

1. Maximum number of comparisons required to find an element.
2. Maximum comparisons if the element is absent.

3. Merge Sort Recurrence

Given the recurrence:

```
[
T(n)=2T(n/2)+n,\quad T(1)=1
]
```

Using the ****Master Theorem****, determine:

- * Time complexity
- * Number of recursion levels when (n=64)

4. Quick Sort Partition

The array is:

...

25 18 40 12 30 15 35

...

Using the **first element as pivot**, perform the **first partition** and determine:

* Final position of the pivot

* Left and right subarrays

5. Fractional Knapsack

A knapsack has capacity **50 kg**.

Item	Weight	Profit
----	-----	-----
A	10	60
B	20	100
C	30	120

Using the **Greedy Algorithm**, calculate:

* Items selected

* Fraction of the last item

* Maximum profit

6. Job Sequencing with Deadlines

Given:

Job	Deadline	Profit
---	-----	-----
J1	2	100
J2	1	19
J3	2	27
J4	1	25
J5	3	15

Find:

- * Optimal job sequence
- * Maximum profit

7. Dijkstra's Algorithm

Find the shortest distance from vertex **A**.

Edge	Weight
----	-----
A-B	4
A-C	2
B-C	5
B-D	10
C-E	3
E-D	4
D-F	11

Determine:

- * Shortest path from A to D
- * Total cost

8. Minimum Spanning Tree (Prim's Algorithm)

For the graph:

Edge	Weight
----	-----
A-B	3
A-D	1
B-D	3
B-C	1
C-D	1
C-E	5
D-E	6

Find:

- * Minimum Spanning Tree
- * Total minimum cost

9. Matrix Chain Multiplication

Matrices:

- * $A_1 = 10 \times 30$

- * $A_2 = 30 \times 5$

- * $A_3 = 5 \times 60$

Calculate:

- * Optimal parenthesization

- * Minimum number of scalar multiplications

10. Longest Common Subsequence (Dynamic Programming)

Given:

X = ABCBDAB

Y = BDCABA

Find:

- * Length of the LCS

- * One possible LCS using Dynamic Programming

Major DAA topics:

- * Time Complexity

- * Recurrence Relations

- * Divide and Conquer

- * Greedy Algorithms

- * Dynamic Programming

- * Graph Algorithms

- * Shortest Path

- * Minimum Spanning Tree

- * Sorting Algorithms

- * Algorithm Analysis